

# Linear algebra: Exercise chapter 9

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1. For the following subspaces  $S$  of  $\mathbb{R}^n$ , find the orthogonal complement  $S^\perp$ :

$$S_1 = \text{span} \left( \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right), \quad S_2 = \text{span} \left( \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right), \quad S_3 = \text{span} \left( \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right), \quad S_4 = \text{span} \left( \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 3 \\ 2 \end{bmatrix} \right).$$

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2. True or false:

- (a) The subspace spanned by  $(1, 1, 0, 0, 0)$  and  $(0, 0, 0, 1, 1)$  is the orthogonal complement of the subspace spanned by  $(1, -1, 0, 0, 0)$  and  $(2, -2, 3, 4, -4)$ .
- (b) Two subspaces that only meet in the zero vector are orthogonal.
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3. Project the vector  $b$  onto the line through  $a$ :

(a)  $b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(b)  $b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, a = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$

Compute the projection matrices for the previous exercise and use these to verify your result.

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4. Project  $b$  onto the column space of  $A$ . Make use of the projection matrix.

(a)  $b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$

(b)  $b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$

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5. Find an orthonormal basis for the space spanned by:

$$a = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, c = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}.$$

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6. Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \end{bmatrix}.$$

Find orthonormal bases for the following spaces:

- Column space  $C(A)$
  - Row space  $C(A^T)$
  - Nullspace  $N(A)$
  - Left nullspace  $N(A^T)$
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7. Consider the least square problem,

$$\min_x \frac{1}{2} \|Ax - b\|^2.$$

For the following data points:

$$\{(-1, 1), (0, 0), (1, 2), (2, 2)\},$$

- Find the closest straight line to them.
  - Compute the error.
  - Check that the error vector is perpendicular to columns of the matrix  $A$ .
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